Dr. Caleb R. Choban

COMPUTATIONAL ASTROPHYSICIST · SULLIVAN PRIZE FELLOW

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Summary

Computational astrophysicist specializing in multidisciplinary scientific research, parallelized galaxy simulation development, simulation forward-modeling, and statistical analysis.

Over seven years of experience developing numerical models, integrated into parallelized simulations, to connect the micro-physics of interstellar dust grains to their galaxy-scale evolution.

Expertise in forward-modeling simulations with radiative transfer codes, generating synthetic photometry and spectra to statistically compare and constrain theoretical models against real datasets.

Experience

PARALLELIZED CODES

- 10 years utilizing supercomputing facilities: Frontera (TACC), Big Red 200 (IU), Expanse (SDSC), Bridges (PSC), Rivanna (UVA).
- Developed and tested physics modules in the **GIZMO** multi-physics, magnetohydrodynamics galaxy simulation code.
- Extensive experience utilizing the parallelized Monte Carlo radiative transfer code **SKIRT**.

FORWARD-MODELING SIMULATIONS

- Produced realistic synthetic photometry and spectra from galaxy simulations with incorporation of point spread function convolution and noise injection.
- Compared synthetic and real observations at various wavelengths and resolutions to constrain theoretical models.

DATA PIPELINES AND STATISTICAL ANALYSIS

- Developed post-processing pipelines to reduce and analyze TB-scale simulation datasets.
- Conducted statistical characterization of synthetic images (brightness distributions, correlations, structure characterization).

Skills & Leadership

Programming	Python · C/C++ · Mathematica · MPI/OpenMP parallelization · HPC · LaTeX · GitHub · PyTorch (novice) · Kokkos parallelization (novice)
Communication	23 science talks · 3 public talks · 7 years teaching physics/astronomy · Numerous public outreach events Extensive experience communicating research to diverse audiences from field experts to the general public.
Research Mentor	Developed and supervised research projects for 2 undergraduates and 2 graduates concluding in 2 scientific manuscripts soon to be published.
REU Director	Directed summer REU for 2 years, reviewing and selecting applicants, coordinating faculty mentors and program logistics, and assessing research outcomes.

Positions

Inaugural Sullivan Prize Fellow — *Indiana University*
FACULTY SPONSOR: SAMIR SALIM — *2023-PRESENT*

Graduate Research Assistant — *University of California San Diego*
ADVISOR: DUŠAN KEREŠ — *2016 - 2023*

Undergraduate Research Assistant — *University of California San Diego*
ADVISOR: ADAM BURGASSER — *2014-2016*

Education

PhD in Physics ADVISED BY DUŠAN KEREŠ — *University of California San Diego (2023)*

MS in Physics — *University of California San Diego (2018)*

BS in Physics w/ Spec. in Astrophysics (Cum Laude) — *University of California San Diego (2015)*

Publications

Corresponding (First) Author

- **“Ashes of FIRE: Modeling Dust Grain Size Evolution in the Local Group with FIRE”**
  C. R. Choban, S. Salim, D. Kereš, J. Roman-Duval, and K. M. Sandstrom, 2026, MNRAS, 549, 1020
- **“A Dusty Dawn: Galactic Dust Buildup at $z \gtrsim 5$”**
  C. R. Choban, S. Salim, D. Kereš, C. C. Hayward, and K. M. Sandstrom, 2025, MNRAS, 537, 1518
- **“A Dusty Locale: Evolution of Galactic Dust Populations from Milky Way to Dwarf-Mass Galaxies”**
  C. R. Choban, D. Kereš, K. M. Sandstrom, P. F. Hopkins, C. C. Hayward, and C. Faucher-Giguère, 2024, MNRAS, 529, 2356
- **“The Galactic Dust-Up: Modeling Dust Evolution in FIRE”**
  C. R. Choban, D. Kereš, P. F. Hopkins, K. M. Sandstrom, C. C. Hayward, and C. Faucher-Giguère, 2022, MNRAS, 514, 4506

Mentee-Led

- **“Non-parametric Galaxy Morphology of FIRE-2 Simulations with a Self-consistent Dust Evolution Model”**
  Alejandro Guzmán-Ortega, **C. R. Choban**, et al., in prep **[draft link]**
- **“Comparing PHANGS Structure Statistics with Simulations to Constrain Feedback Models”**
  Hao-Sheng Wang, **C. R. Choban**, et al., in prep **[draft link]**

Co-Author

- **”Observing Co-Located Neutral and Ionized Gas-Phase Iron Depletion in the Magellanic Clouds”**
  Yun Qi Li, Jessica K. Werk, **Caleb R. Choban**, Julia Roman-Duval, Kirill Tchernyshyov, J. Xavier Prochaska, Doyeon A. Kim, Arianna S. Long, submitted to ApJ
- **“Fire-3: Updated Stellar Evolution Models, Yields, & Microphysics and Fitting Functions for Applications in Galaxy Simulations”**
  P. F. Hopkins, A. Wetzel, C. Wheeler, R. Sanderson, M. Y. Grudić, O. Sameie, M. Boylan-Kolchin, M. Orr, X. Ma, C. Faucher-Giguère, D. Kereš, E. Quataert, K. Su, J. Moreno, R. Feldmann, J. S. Bullock, S. R. Loebman, D. Anglés-Alcázar, J. Stern, L. Necib, **C. R. Choban**, C. C. Hayward, 2023, MNRAS, 519, 3154
- **“Characterization of the Very Low Mass Secondary in the GJ 660.1AB System”**
  C. Aganze, A. J. Burgasser, J. K. Faherty, **C. R. Choban**, I. Escala, M. A. Lopez, Y. Jin, T. Tamiya, M. Tallis, W. Rockward, 2016, ApJ, 151, 6A
- **“The First Brown Dwarf in the 32 Orionis Association”**
  A. J. Burgasser, M. A. Lopez , E. E. Mamajek, J. Gagne , J. K. Faherty, M. Tallis, **C. R. Choban**, I. Escala, C. Aganze, 2016, ApJ, 820, 32B

Invited Talks

Exploring the Aromatic Universe in the JWST era	*Western University (2026)*
Physics and Astronomy Colloquium	*University of Toledo (2026)*
Kavli Summer Program in Astrophysics	*University of Virginia (2025)*
Star Formation & ISM Seminar	*Center for Computational Astro. (2025)*
Astrophysics Seminar	*Purdue University (2025)*
Dust and Gas throughout Cosmic Time	*Hiroshima, Japan (2024)*
Building Physical Simulations of PAH Emission in Galaxies II Workshop	*University of Virginia (2024)*
Journal Club	*UI Urbana-Champaign (2024)*
Astronomy Seminar	*University of Notre Dame (2024)*
Friday Lunch Time Astrophysics Seminar	*UC Santa Cruz (2023)*
Star Formation/ISM Rendezvous Seminar	*Princeton (2022)*
FIRE Collaboration Seminar	*Virtual (2021)*

Community Engagement

Mentorship/Leadership

Alice Palma Undergraduate Program	*Organizer and Mentor (2023-26)*
Preparing the Emerging Next Generation Using Inclusive Networking (PENGUIN)	*Mentor (2026)*
Kavli Summer Program in Astrophysics	*Mentor (2025)*
PyIU: Python Crash Course and Workshop for Scientists	*Organizer and Presenter (2024-26)*
Graduate/Undergraduate Peer-to-Peer Mentoring Program	*Founder and Organizer (2019-23)*
Physics Graduate Council	*Representative (2018-23)*
Society of Physics Students	*Vice President (2014-16)*

Public Speaking

Hutton Honors Council Association *JWST: Our Dusty Universe* *(2024)*
Science on Screen: Buskirk-Chumley Theater *Galaxy Movie Theater* *(2024)*
Eclipsed by Chocolate: WonderLab Museum of Science *The Science of Solar Eclipses* *(2024)*

Science Outreach

Solar Eclipse IU *(2024)*
San Diego Air & Space Museum Space Day *(2022-23)*
Transfer Student STEM Research Seminar *(2021)*
UCSD Young Physicists Program *(2018-20)*
San Diego Festival for Science and Engineering *(2015-20)*
Southern California Physics Graduate Admissions Bootcamp *(2016-17)*
Institute of the Americas Science & Innovation Camp *(2015-17)*
Intertribal Youth: Indigenize Education *(2015-17)*

Honors, Awards, & Grants

Poster Prize @ Dust and Gas throughout Cosmic Time *2024*
Distinguished Senior Graduate Teaching Award *2023*
AAS International Travel Grant *2023*
UCSD Graduate Student Association Travel Grant *2019*
NSF Graduate Research Fellowship, Honorable Mention *2018*
UCSD Physical Sciences Dean's Undergraduate Award for Excellence *2015-16*
Provost Honours, UCSD *2013-15*

Teaching Experience

Courses

Mathematical and Computational Physics
Laboratory Projects w/ Arduinos
Circuits and Electronics Lab
Physics Lab E&M, Waves, & Optics
Electricity and Magnetism
Observational Astrophysics Research Lab
Galaxies and Quasars
Cosmology
Life in the Universe
Galaxies and Cosmology

Workshops

Using AI in Research
Intro to Git and GitHub
Intro to PyTorch
Coding Design
How to Build an Academic Website
Data Structures and Advanced Plotting w/ Python
Time Management